Bisoprolol Clinical Data Analysis

Project Overview Report

Data Mining Approaches in Clinical Trial Analysis

Parameter	Detail
Drug Analyzed	Bisoprolol (beta-blocker)
Indication	Hypertension & elevated heart rate
Dose Range	2.5 mg, 5 mg, 10 mg
Patient Age Range	45 – 75 years (mean: 60)
Approach	Exploratory data analysis + predictive ML

Project Purpose

This project demonstrates how data mining and machine learning techniques can be applied to clinical trial data. The analysis investigates the dose-response relationship of Bisoprolol, a commonly prescribed beta-blocker, on three key cardiovascular metrics: systolic blood pressure, diastolic blood pressure, and resting heart rate.

Data was collected from a cohort of patients aged 45–75 years, with baseline measurements taken prior to any dosing and follow-up measurements recorded at 2.5 mg, 5 mg, and 10 mg doses.

Key Findings

1. Dose-Response Relationship

Analysis confirmed a consistent, statistically meaningful inverse relationship between Bisoprolol dose and all three cardiovascular measures:

- Heart rate, systolic BP, and diastolic BP all decreased as dose increased from 0 to 10 mg.
- Mean-trend line plots and correlation coefficients confirmed negative linear trends across all metrics.
- A dose of 10 mg was sufficient to bring most patients into a clinically acceptable blood pressure range.

2. Blood Pressure Normalization

Boxplot analysis across dose levels showed:

- **Before treatment, most patients exceeded the 140 mmHg hypertension threshold. By 10 mg, mean systolic BP fell toward the 120 mmHg normal reference line.**
- **Similar reduction observed — mean diastolic values converged near the 80 mmHg normal range at the 10 mg dose.**
- **A steady decline in resting heart rate was seen across all dose increments.**

Methodology

Exploratory Data Analysis

The initial phase employed visualization-driven EDA to characterize the dataset:

- Age distribution histogram revealed a roughly uniform cohort between 45 and 75, with a mean of 60 years.
- Boxplots per dose level provided a view of spread, median, and outliers across all cardiovascular measurements.
- Mean trend lines overlaid on boxplots made dose-response trajectories immediately visible.

Linear Regression Analysis

Scikit-learn's LinearRegression was used to model dose as an independent variable against mean systolic BP, diastolic BP, and heart rate. Pearson correlation coefficients were computed to quantify the strength of the dose-response relationship.

Predictive ML Model

A second-stage model was trained to predict the change in blood pressure at a given dose, taking a patient's baseline systolic and diastolic values plus the intended dose as input features.

- Dataset construction: each patient contributed three training samples — one per dose level — with the target being the absolute change from baseline.
- Train/test split: 75% training, 25% test (random state = 42).
- Separate models were trained for systolic and diastolic change.

Model Performance

The trained models were evaluated using Mean Absolute Error (MAE) and R-squared (R2):

Metric	Systolic Model	Diastolic Model
Evaluation Metric	MAE & R2 Score	MAE & R2 Score
Performance	High predictive accuracy	High predictive accuracy
Author's Assessment	"Model is very good at predicting"	Consistent with systolic results

Interactive Predictor

The final section of the notebook implements a user-facing blood pressure predictor. A clinician or researcher can enter a patient's baseline systolic and diastolic BP and a target dose (2.5, 5, or 10 mg). The models then output:

- Predicted systolic and diastolic change from baseline.
- Estimated final systolic and diastolic blood pressure post-treatment.
- Link to the streamlit application: **https://blood-pressure-predictor-ghztxydshe7qt6a6cbphis.streamlit.app**
- Streamlit application source code: **https://github.com/shaxadaily/blood-pressure-predictor**

Note: When you access the streamlit application, you need to upload .csv file (bisoprolol_final.csv), which you can find in my source code.

Conclusion

The Bisoprolol project successfully demonstrates that data mining and regression-based machine learning can extract clinically meaningful insights from cardiovascular trial data. The dose-response patterns are clearly captured, and the predictive model provides a practical tool for estimating treatment outcomes based on individual patient baselines.

The work highlights the value of combining exploratory visualization with supervised learning in clinical research, enabling both population-level understanding and individualized prediction.